

Logarithmic Functions & Properties

ANSWER KEY



Converting between exponential and logarithmic form

- 1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

$$2^5 = 32 \Leftrightarrow \log_2 32 = 5. \log_3 81 = 4 \Leftrightarrow 3^4 = 81.$$

- 2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.

$$\log_5 125 = 3 \text{ since } 5^3 = 125. \log_2 \frac{1}{8} = -3 \text{ since } 2^{-3} = \frac{1}{8}.$$

Properties of logarithms

- 3 Use logarithm properties to expand $\log \left(\frac{x^3 \sqrt{y}}{z^2} \right)$ fully.

$$\log \left(\frac{x^3 \sqrt{y}}{z^2} \right) = \log(x^3) + \log(y^{1/2}) - \log(z^2) = 3\log x + \frac{1}{2}\log y - 2\log z.$$

- 4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

$$2\log x - \frac{1}{3}\log y + \log z = \log(x^2) - \log(y^{1/3}) + \log z = \log \left(\frac{x^2 z}{y^{1/3}} \right).$$

- 5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

$$\log_7 50 = \frac{\ln 50}{\ln 7} = \frac{3.912}{1.946} \approx 2.010.$$

Visualizing logarithmic functions

- 6 The graph shows $f(x) = \ln(x)$, the natural logarithm.

a State the domain and the vertical asymptote shown in the graph. Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \rightarrow 0^+$.

b State the x -intercept shown. x -intercept at $(1, 0)$, since $\ln(1) = 0$.

Solving exponential equations using logarithms

- 7 Solve $5^x = 40$ for x , to three decimal places.

$$\text{Taking log of both sides: } x\log 5 = \log 40, \text{ so } x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292.$$

- 8 Solve $3(2)^{x-1} = 45$ for x , to three decimal places.

$$2^{x-1} = 15. \text{ Taking log: } (x-1)\log 2 = \log 15, \text{ so } x-1 = \frac{\log 15}{\log 2} \approx 3.907, \text{ giving } x \approx 4.907.$$

Solving logarithmic equations

- 9 Solve $\log_2(x + 3) = 4$ for x , and check that the solution keeps the argument positive.

$x + 3 = 2^4 = 16$, so $x = 13$. Check: $x + 3 = 16 > 0$ ✓. Solution: $x = 13$.

- 10 Solve $\log(x) + \log(x - 3) = 1$ for x , checking both solutions against the domain.

Combine: $\log[x(x - 3)] = 1$, so $x(x - 3) = 10$: $x^2 - 3x - 10 = 0$, $(x - 5)(x + 2) = 0$: $x = 5$ or $x = -2$.
Since $x > 0$ and $x - 3 > 0$ are required (domain of the original logs), $x = -2$ is rejected. Solution: $x = 5$.

Applications: the pH scale and decibels

- 11 The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.

$\text{pH} = -\log(3.2 \times 10^{-5}) \approx -(-4.495) = 4.49$.

- 12 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

Difference $= 10\log(1000) - 10\log(1) = 10\log(1000) = 10(3) = 30$ decibels.

Transformations of logarithmic functions

- 13 Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.

$g(x) = \ln(x + 2) - 3$ shifts $f(x) = \ln(x)$ left 2 units and down 3 units. The vertical asymptote shifts from $x = 0$ to $x = -2$, and the domain becomes $x > -2$.

- 14 Find the domain of $h(x) = \log(4 - x^2)$.

Require $4 - x^2 > 0$: $x^2 < 4$, so $-2 < x < 2$. Domain: $(-2, 2)$.

Solving equations that combine exponentials and logarithms

- 15 Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.

Let $u = e^x$: $u^2 - 5u + 6 = 0$, so $(u - 2)(u - 3) = 0$, giving $u = 2$ or $u = 3$. Since $u = e^x > 0$ for all real x , both are valid: $e^x = 2 \Rightarrow x = \ln 2$, or $e^x = 3 \Rightarrow x = \ln 3$.

- 16 Solve $\log_2(x) + \log_2(x - 2) = 3$ for x , checking the domain.

Combine: $\log_2[x(x - 2)] = 3$, so $x(x - 2) = 2^3 = 8$: $x^2 - 2x - 8 = 0$, $(x - 4)(x + 2) = 0$: $x = 4$ or $x = -2$. Domain requires $x > 0$ and $x - 2 > 0$ (i.e. $x > 2$), so $x = -2$ is rejected. Solution: $x = 4$.

Comparing logarithmic growth to other function families

- 17 Explain why logarithmic functions grow much more slowly than linear functions for large x , and give a numerical comparison of $\log(x)$ and x at $x = 1000$.

Logarithms compress large inputs: each tenfold increase in x adds only 1 to $\log(x)$, while a linear function adds proportionally to the increase in x itself. At $x = 1000$: $\log(1000) = 3$, while the linear function $y = x$ gives 1000 — the logarithm is vastly smaller, illustrating its extremely slow long-run growth.

Modeling with logarithms: the Richter scale

- 18 Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

$7 - 5 = \log\left(\frac{I_7}{I_0}\right) - \log\left(\frac{I_5}{I_0}\right) = \log\left(\frac{I_7}{I_5}\right) = 2$, so $\frac{I_7}{I_5} = 10^2 = 100$. The magnitude-7 earthquake is 100 times more intense.

- 19 If an earthquake's intensity is 10,000 times a reference intensity I_0 , find its Richter magnitude.

$$M = \log(10000) = \log(10^4) = 4.$$

Multiple choice — easy 15

- 20 Evaluate $\log_2(2)$ without a calculator.

- a. 1 ✓
- b. -1
- c. 0
- d. 2

$\log_2(2) = 1$ since $2^1 = 2$.

21 Evaluate $\log_9(81)$ without a calculator.

a. 3

b. 1

c. -2

d. 2 ✓

$\log_9(81) = 2$ since $9^2 = 81$.

22 Evaluate $\log_6(36)$ without a calculator.

a. 2 ✓

b. 1

c. -2

d. 3

$\log_6(36) = 2$ since $6^2 = 36$.

23 Evaluate $\log_2(8)$ without a calculator.

a. 2

b. 4

c. -3

d. 3 ✓

$\log_2(8) = 3$ since $2^3 = 8$.

24 Evaluate $\log_4(16)$ without a calculator.

a. 3

b. 2 ✓

c. -2

d. 1

$\log_4(16) = 2$ since $4^2 = 16$.

25 Evaluate $\log_4 \left(\frac{1}{64} \right)$ without a calculator.

a. -3 ✓

b. -2

c. 3

d. -4

$$\log_4 \left(\frac{1}{64} \right) = -3 \text{ since } 4^{-3} = \frac{1}{64}.$$

26 Evaluate $\log_4 (64)$ without a calculator.

a. -3

b. 3 ✓

c. 4

d. 2

$$\log_4 (64) = 3 \text{ since } 4^3 = 64.$$

27 Evaluate $\log_3 \left(\frac{1}{9} \right)$ without a calculator.

a. -1

b. 2

c. -3

d. -2 ✓

$$\log_3 \left(\frac{1}{9} \right) = -2 \text{ since } 3^{-2} = \frac{1}{9}.$$

28 Evaluate $\log_6\left(\frac{1}{216}\right)$ without a calculator.

a. 3

b. -4

c. -3 ✓

d. -2

$$\log_6\left(\frac{1}{216}\right) = -3 \text{ since } 6^{-3} = \frac{1}{216}.$$

29 Evaluate $\log_4(4)$ without a calculator.

a. 2

b. 0

c. -1

d. 1 ✓

$$\log_4(4) = 1 \text{ since } 4^1 = 4.$$

30 Evaluate $\log_3\left(\frac{1}{3}\right)$ without a calculator.

a. -2

b. 0

c. 1

d. -1 ✓

$$\log_3\left(\frac{1}{3}\right) = -1 \text{ since } 3^{-1} = \frac{1}{3}.$$

31 Evaluate $\log_2\left(\frac{1}{2}\right)$ without a calculator.

a. 0

b. 1

c. -2

d. -1 ✓

$$\log_2\left(\frac{1}{2}\right) = -1 \text{ since } 2^{-1} = \frac{1}{2}.$$

32 Evaluate $\log_5(5)$ without a calculator.

- a. 0
- b. 2
- c. 1 ✓
- d. -1

$\log_5(5) = 1$ since $5^1 = 5$.

33 Evaluate $\log_4\left(\frac{1}{256}\right)$ without a calculator.

- a. -3
- b. -5
- c. -4 ✓
- d. 4

$\log_4\left(\frac{1}{256}\right) = -4$ since $4^{-4} = \frac{1}{256}$.

34 Evaluate $\log_5(625)$ without a calculator.

- a. 4 ✓
- b. 5
- c. 3
- d. -4

$\log_5(625) = 4$ since $5^4 = 625$.

Multiple choice — medium 15

35 Solve $\log_6(x) = 3$ for x .

a. 215

b. 217

c. 216 ✓

d. 218

$$x = 6^3 = 216, \text{ so } x = 216.$$

36 Solve $\log_4(x - 6) = 1$ for x .

a. -2

b. 11

c. 10 ✓

d. 9

$$x - 6 = 4^1 = 4, \text{ so } x = 4 + 6 = 10.$$

37 Solve $\log_2(x + 9) = 2$ for x .

a. 13

b. -4

c. -5 ✓

d. -6

$$x + 9 = 2^2 = 4, \text{ so } x = 4 - 9 = -5.$$

38 Solve $\log_5(x - 5) = 3$ for x .

a. 129

b. 130 ✓

c. 120

d. 131

$$x - 5 = 5^3 = 125, \text{ so } x = 125 + 5 = 130.$$

39 Solve $\log_3(x - 6) = 2$ for x .

a. 15 ✓

b. 3

c. 14

d. 16

$$x - 6 = 3^2 = 9, \text{ so } x = 9 + 6 = 15.$$

40 Solve $\log_4(x + 9) = 1$ for x .

a. 13

b. -4

c. -5 ✓

d. -6

$$x + 9 = 4^1 = 4, \text{ so } x = 4 - 9 = -5.$$

41 Solve $\log_6(x + 6) = 1$ for x .

a. -1

b. 1

c. 0 ✓

d. 12

$$x + 6 = 6^1 = 6, \text{ so } x = 6 - 6 = 0.$$

42 Solve $\log_3(x + 4) = 2$ for x .

a. 13

b. 4

c. 5 ✓

d. 6

$$x + 4 = 3^2 = 9, \text{ so } x = 9 - 4 = 5.$$

43 Solve $\log_4(x + 7) = 1$ for x .

a. -2

b. -3 ✓

c. -4

d. 11

$$x + 7 = 4^1 = 4, \text{ so } x = 4 - 7 = -3.$$

44 Solve $\log_5(x + 3) = 3$ for x .

a. 123

b. 122 ✓

c. 128

d. 121

$$x + 3 = 5^3 = 125, \text{ so } x = 125 - 3 = 122.$$

45 Solve $\log_6(x - 3) = 1$ for x .

a. 8

b. 9 ✓

c. 10

d. 3

$$x - 3 = 6^1 = 6, \text{ so } x = 6 + 3 = 9.$$

46 Solve $\log_6(x + 2) = 2$ for x .

a. 33

b. 34 ✓

c. 35

d. 38

$$x + 2 = 6^2 = 36, \text{ so } x = 36 - 2 = 34.$$

47 Solve $\log_2(x + 7) = 1$ for x .

a. -5 ✓

b. 9

c. -4

d. -6

$$x + 7 = 2^1 = 2, \text{ so } x = 2 - 7 = -5.$$

48 Solve $\log_5(x - 8) = 2$ for x .

a. 34

b. 17

c. 32

d. 33 ✓

$$x - 8 = 5^2 = 25, \text{ so } x = 25 + 8 = 33.$$

49 Solve $\log_6(x - 6) = 2$ for x .

a. 42 ✓

b. 43

c. 30

d. 41

$x - 6 = 6^2 = 36$, so $x = 36 + 6 = 42$.

Multiple choice — hard 15

50 Solve $4^x = 138$ for x , to three decimal places.

a. 4.054

b. 3.654

c. 3.554 ✓

d. 3.454

Taking log of both sides: $x \log 4 = \log 138$, so $x = \frac{\log 138}{\log 4} \approx 3.554$.

51 Solve $9^x = 165$ for x , to three decimal places.

a. 2.824

b. 2.424

c. 2.224

d. 2.324 ✓

Taking log of both sides: $x \log 9 = \log 165$, so $x = \frac{\log 165}{\log 9} \approx 2.324$.

52 Solve $9^x = 96$ for x , to three decimal places.

a. 2.177

b. 2.077 ✓

c. 1.977

d. 2.577

Taking log of both sides: $x \log 9 = \log 96$, so $x = \frac{\log 96}{\log 9} \approx 2.077$.

53 Solve $2^x = 109$ for x , to three decimal places.

a. 6.868

b. 7.268

c. 6.668

d. 6.768 ✓

Taking log of both sides: $x \log 2 = \log 109$, so $x = \frac{\log 109}{\log 2} \approx 6.768$.

54 Solve $6^x = 90$ for x , to three decimal places.

a. 2.411

b. 2.511 ✓

c. 2.611

d. 3.011

Taking log of both sides: $x \log 6 = \log 90$, so $x = \frac{\log 90}{\log 6} \approx 2.511$.

- 55 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10 times more intense than another, find the difference in their decibel levels.
- a. 30
 - b. 20
 - c. 10 ✓
 - d. 1

Difference = $10\log(10) - 10\log(1) = 10(1) = 10$ decibels.

- 56 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^6 times more intense than another, find the difference in their decibel levels.
- a. 50
 - b. 70
 - c. 60 ✓
 - d. 6

Difference = $10\log(10^6) - 10\log(1) = 10(6) = 60$ decibels.

- 57 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^3 times more intense than another, find the difference in their decibel levels.
- a. 40
 - b. 20
 - c. 3
 - d. 30 ✓

Difference = $10\log(10^3) - 10\log(1) = 10(3) = 30$ decibels.

- 58 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^2 times more intense than another, find the difference in their decibel levels.

a. 20 ✓

b. 10

c. 2

d. 30

Difference = $10\log(10^2) - 10\log(1) = 10(2) = 20$ decibels.

- 59 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^5 times more intense than another, find the difference in their decibel levels.

a. 5

b. 40

c. 60

d. 50 ✓

Difference = $10\log(10^5) - 10\log(1) = 10(5) = 50$ decibels.

- 60 Solve $\log_{40}(x) + \log_{40}(x + 3) = 1$ for x , checking both solutions against the domain.

a. -3

b. 6

c. 5 ✓

d. -8

Combine: $\log_{40}[x(x + 3)] = 1$, so $x(x + 3) = 40$: $x^2 + 3x - 40 = 0$, $(x - 5)(x - (-8)) = 0$: $x = 5$ or $x = -8$. Since $x > 0$ and $x + 3 > 0$ are required, $x = -8$ is rejected. Solution: $x = 5$.

61 Solve $\log_{21}(x) + \log_{21}(x - 4) = 1$ for x , checking both solutions against the domain.

- a. 8
- b. -3
- c. 7 ✓
- d. 4

Combine: $\log_{21}[x(x - 4)] = 1$, so $x(x - 4) = 21$: $x^2 - 4x - 21 = 0$, $(x - 7)(x - (-3)) = 0$: $x = 7$ or $x = -3$. Since $x > 0$ and $x - 4 > 0$ are required, $x = -3$ is rejected. Solution: $x = 7$.

62 Solve $\log_{12}(x) + \log_{12}(x - 4) = 1$ for x , checking both solutions against the domain.

- a. 4
- b. -2
- c. 7
- d. 6 ✓

Combine: $\log_{12}[x(x - 4)] = 1$, so $x(x - 4) = 12$: $x^2 - 4x - 12 = 0$, $(x - 6)(x - (-2)) = 0$: $x = 6$ or $x = -2$. Since $x > 0$ and $x - 4 > 0$ are required, $x = -2$ is rejected. Solution: $x = 6$.

63 Solve $\log_6(x) + \log_6(x - 5) = 1$ for x , checking both solutions against the domain.

- a. 6 ✓
- b. 7
- c. -1
- d. 5

Combine: $\log_6[x(x - 5)] = 1$, so $x(x - 5) = 6$: $x^2 - 5x - 6 = 0$, $(x - 6)(x - (-1)) = 0$: $x = 6$ or $x = -1$. Since $x > 0$ and $x - 5 > 0$ are required, $x = -1$ is rejected. Solution: $x = 6$.

64 Solve $\log_{18}(x) + \log_{18}(x + 7) = 1$ for x , checking both solutions against the domain.

a. 3

b. 2 ✓

c. -9

d. -7

Combine: $\log_{18}[x(x + 7)] = 1$, so $x(x + 7) = 18$: $x^2 + 7x - 18 = 0$, $(x - 2)(x - (-9)) = 0$: $x = 2$ or $x = -9$. Since $x > 0$ and $x + 7 > 0$ are required, $x = -9$ is rejected. Solution: $x = 2$.